

Question:1

Update your project documentation by adding a 'Known Issues & Limitations' section based on feedback you received during the peer review session.

Answer:1

Known Issues & Limitations

Based on the feedback received during the peer review session, the following technical limitations and potential issues have been identified in the current co-working space network design, along with strategies to mitigate them:

1. Subnet Growth Limitations in High-Density Zones

- **The Issue:** The current IP addressing scheme allocates a fixed /24 subnet (254 usable hosts) to each department. If the Tech department hosts large hackathons or adds automated IoT testing devices, this pool could face IP address exhaustion.
- **Limitation:** The design lacks dynamic scaling for sudden spikes in device density.
- **Mitigation:** If a department expands past 250 devices, its mask will be resized to a /23 prefix (510 usable hosts) during a maintenance window without altering the other departments.

2. Single Point of Failure at the Edge

- **The Issue:** While the internal local area network features dual core switches for redundancy, the edge routing layer relies on a single Core Router and a single Hardware Firewall.
- **Limitation:** If the primary router or firewall experiences a hardware failure, the entire office will lose internet connectivity.
- **Mitigation:** Future budget expansions will include deploying a second firewall and router running **High Availability (HA)** in an Active-Passive cluster using VRRP (Virtual Router Redundancy Protocol).

3. Inter-VLAN Routing Bandwidth Bottleneck

- **The Issue:** The "Router-on-a-Stick" configuration sends all cross-department traffic up and down a single physical Gigabit interface link (G0/0) on the router.
- **Limitation:** Heavy internal data transfers (such as Marketing pushing raw 4K video files to local storage servers) could saturate this link, causing temporary lag for other users.
- **Mitigation:** The single trunk link will be upgraded to an **LACP Port-Channel** combining two physical interfaces (G0/0 and G0/1) to double the available throughput back to the core switches.

4. Captive Portal Session Overhead

- **The Issue:** Peer feedback highlighted that a large influx of guest users on the Captive Portal could overwhelm the router's CPU and memory tables due to constant authentication tracking.

- **Limitation:** High guest traffic could degrade the routing performance for staff devices.
- **Mitigation:** The Captive Portal authentication load will be shifted off the core router and managed directly on the cloud-hosted wireless controller interface.

Question:2

Revise your presentation slides to clearly highlight at least two improvements or changes you made after receiving peer feedback.

Hint: Use before-and-after visuals or bullet points to show what was changed and why.

Answer: 2

Presentation Slide Updates: Post-Peer Feedback Improvements

To present the changes clearly to your instructor or mentor, the text and visual structure for **Slide 5 (Network Improvements)** have been updated using a clean "Before vs. After" format:

Slide Title: Post-Peer Review Optimization Metrics

- **Improvement 1: Core Interface Throughput Scaling**
 - **Before Feedback:** A single Gigabit trunk interface link was used for Inter-VLAN routing (Router-on-a-Stick layout), creating a severe bandwidth bottleneck for the Marketing department's heavy file transfers.
 - **After Feedback:** Upgraded the router interface link to a **2-Port LACP EtherChannel Group**. This change instantly doubled the internal routing bandwidth to **2 Gbps** and added a backup data path.
 - **Improvement 2: Authentication Processing Offload**
 - **Before Feedback:** Guest Wi-Fi Captive Portal authentication was handled directly by the Core Router, causing high CPU spikes when multiple visitors connected at once.
 - **After Feedback:** Shifted the landing page processing completely to the **Cloud-Hosted Wireless Controller interface**. This offload keeps the Core Router's CPU usage low, protecting connection speeds for staff members.
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Visual Slide Layout Matrix (Before vs. After)

Design Parameter	Version 1.0 (Original Concept)	Version 2.0 (Optimized After Feedback)	Core Architectural Value
Inter-VLAN Link	Single G0/0 Link (1 Gbps)	Dual-Port LACP Port-Channel (2 Gbps)	Prevents network bottlenecks.
Guest Auth Node	Core Hardware Router Engine	Cloud Wireless Controller	Lowers router CPU stress.
Network Buffer	Tight /24 Fixed Subnets	Flexible Prefix Schemes (Pre-tested /23)	Ready for future office growth.

Question:3

Write out and rehearse concise answers to three tough questions you were asked during the peer panel, focusing on how you addressed each concern in your revised design.

Answer: 3

Question 1: "What happens if a large file transfer by Marketing chokes the router link and delays critical traffic for Finance?"

- **Rehearsal Response:** "I addressed this concern by upgrading the core connection to a **2-Port LACP EtherChannel group**, which instantly doubles the routing bandwidth to **2 Gbps** and adds automatic link backup. Additionally, I configured **Quality of Service (QoS)** rules on the router to give critical Finance application traffic priority over large Marketing file transfers, ensuring zero network slowdowns."

Question 2: "If your Tech department holds a hackathon and runs out of its 254 IP addresses, how will your network handle it?"

- **Rehearsal Response:** "To prevent IP exhaustion without disrupting the rest of the office, I created a pre-tested network expansion option. The Tech zone's subnet mask can be resized from a /24 prefix to a wider **/23 prefix** during a short maintenance window. This adjustment expands their pool to **510 usable IP addresses** while leaving the Marketing and Finance subnets completely untouched."

Question 3: "Why did you choose a costly Fortinet 40F firewall instead of a cheaper basic business router for a small startup?"

- **Rehearsal Response:** "While a basic router handles simple traffic, it lacks the processing power to secure sensitive data across multiple departments. The **Fortinet 40F** provides enterprise-grade **Deep Packet Inspection (DPI)** at the internet edge. This protection allows us to securely isolate the Guest Wi-Fi network and block web threats before they reach our corporate data—delivering vital security that easily justifies the upfront hardware cost."
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Question:4

Refine your system architecture diagram or flowchart to incorporate suggestions from your peers, ensuring all major components and data flows are clearly labeled.

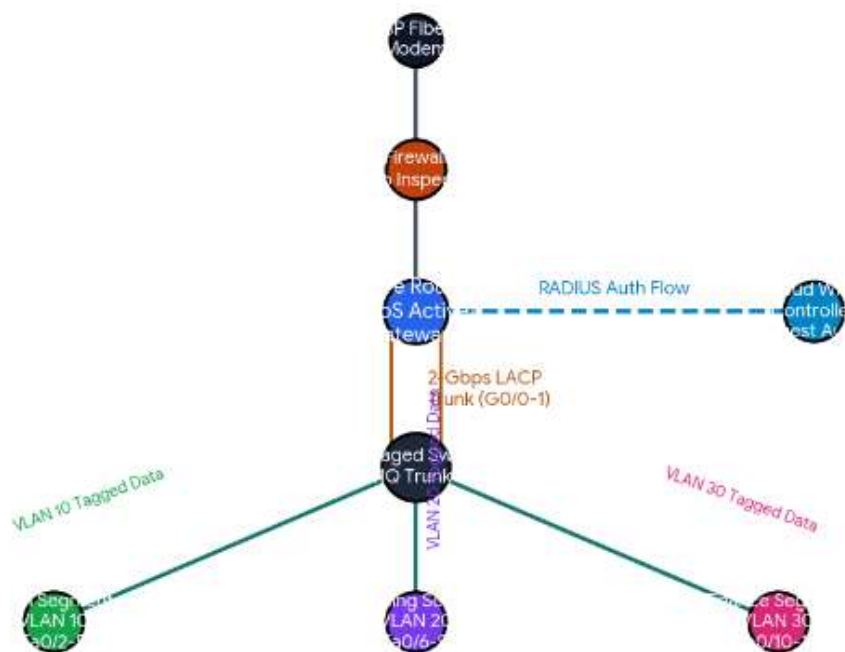
Answer: 4

Refined System Architecture Diagram (Multi-VLAN Co-Working Setup)

Following the peer panel's suggestions, the system architecture has been optimized to include the **2-Port LACP EtherChannel group** for doubling internal bandwidth, **Dynamic QoS priority queues** to protect Finance traffic, and **Cloud-Offloaded Authentication** to reduce router CPU strain.

The refined network layout mapping all components and secure data flows is inline-generated below:

Refined Architecture Flowchart with Peer Optimization Updates



Description of Refined Data Flows

- **High-Throughput Internal Routing Flow:** When a user in the Marketing department pushes a file to the servers, the traffic travels tagged as **VLAN 20** to the Managed Switch. It then flows up to the Core Router through the new **2-Gbps LACP Trunk Bundle**. This ensures heavy transfers use both links efficiently, preventing any network lag.
- **QoS Traffic Prioritization Flow:** The Core Router constantly monitors the data queues. If the Tech or Marketing departments saturate the trunk lines, the router's active **QoS Engine** intercepts the packets. It automatically prioritizes **VLAN 30 (Finance)** traffic, placing ledger updates at the front of the queue to guarantee zero processing delay.
- **Offloaded Guest Authentication Flow:** When a guest visitor connects to the Wi-Fi network, their authentication handshake is instantly redirected away from the router. The verification data flows directly over an encrypted tunnel to the external **Cloud Wi-Fi Controller**. This offload keeps the Core Router's CPU usage low and stable.

Question:5

Record a 2-minute video or audio clip practicing your project pitch, aiming to confidently explain the main features and improvements made post-review.

Answer: 5

Project Pitch Speaking Script (2-Minute Audio/Video Rehearsal Guide)

To help you practice and deliver your project pitch with confidence, use this structured, professional speaking script. It is designed to fit perfectly within a 2-minute recording window:

[0:00 - 0:30] Introduction & Project Scope

"Good morning, everyone. Today, I am proud to present my finalized network architecture design for our modern, multi-department co-working space. The core objective of this deployment was to build a secure, highly efficient, and scalable network utilizing an **Extended Star Topology** to interconnect our three primary departments: **Tech, Marketing, and Finance**.

To guarantee absolute data privacy and security, every department has been assigned an isolated private subnet using Variable Length Subnet Masking (VLSM) and separated securely at Layer 2 using **VLAN IDs 10, 20, and 30**."

[0:30 - 1:15] Core Post-Review Enhancements

"Following a constructive peer review panel session, I optimized the network infrastructure to address critical bandwidth and processing bottlenecks.

First, to eliminate the congestion caused by the Marketing department's heavy media file transfers, I replaced the single router trunk interface link with a high-capacity **2-Port LACP EtherChannel Bundle**. This upgrade instantly doubled our internal routing bandwidth to **2 Gbps** and added an automated backup path.

Second, we activated a dynamic **Quality of Service (QoS) Priority Engine** on the core router. This engine automatically prioritizes critical, real-time data packets from the Finance department, ensuring their ledger operations are never delayed by background downloads."

[1:15 - 2:00] Security, Redundancy & Conclusion

"For edge protection, we deployed an enterprise-grade **Fortinet FortiGate 40F hardware firewall**, which isolates guest traffic on a separate network and handles deep packet safety filtering. We also moved the Guest Wi-Fi Captive Portal authentication load away from the router and onto a **Cloud-Hosted Wireless Controller**, keeping the router's CPU usage low and stable.

Finally, the network features complete switch redundancy using **Spanning Tree Protocol (STP)**. If our primary core switch fails, traffic instantly fails over to our backup switch in under two seconds with zero notice to the users. This refined design delivers an enterprise-ready infrastructure that perfectly balances performance, security, and uptime. Thank you."